

Smart City · Premium Course

Full-day camp (5 hrs) · 10–14 yrs · from KES 5,000

Premium course outline **PREMIUM**

Overview

Teams of 3–4 each build one city subsystem, then connect them on a shared city board.

DETAIL

Track	Arduino Robotics & Electronics
Format	Full-day camp (5 hrs)
Ages	10–14 yrs
Level	Intermediate
Price	from KES 5,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Kits are lent during sessions; buy one to keep building at home.
Standalone	Yes
Prerequisites	None — standalone

Programme

1. **Street Lights.** Photoresistor-activated LEDs that turn on at "night".
2. **Traffic System.** Multi-intersection traffic lights with a pedestrian crossing button.
3. **Parking Garage.** Ultrasonic sensor counts cars in/out; an LED sign shows FULL/ OPEN.
4. **Weather Station.** DHT11 + LDR display conditions on an LCD.

What they take home



Their subsystem module + a group photo of the full city

Why parents love it

“Kids collaborate like real city engineers — each team owns a piece, then they integrate it all.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Smart City**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.